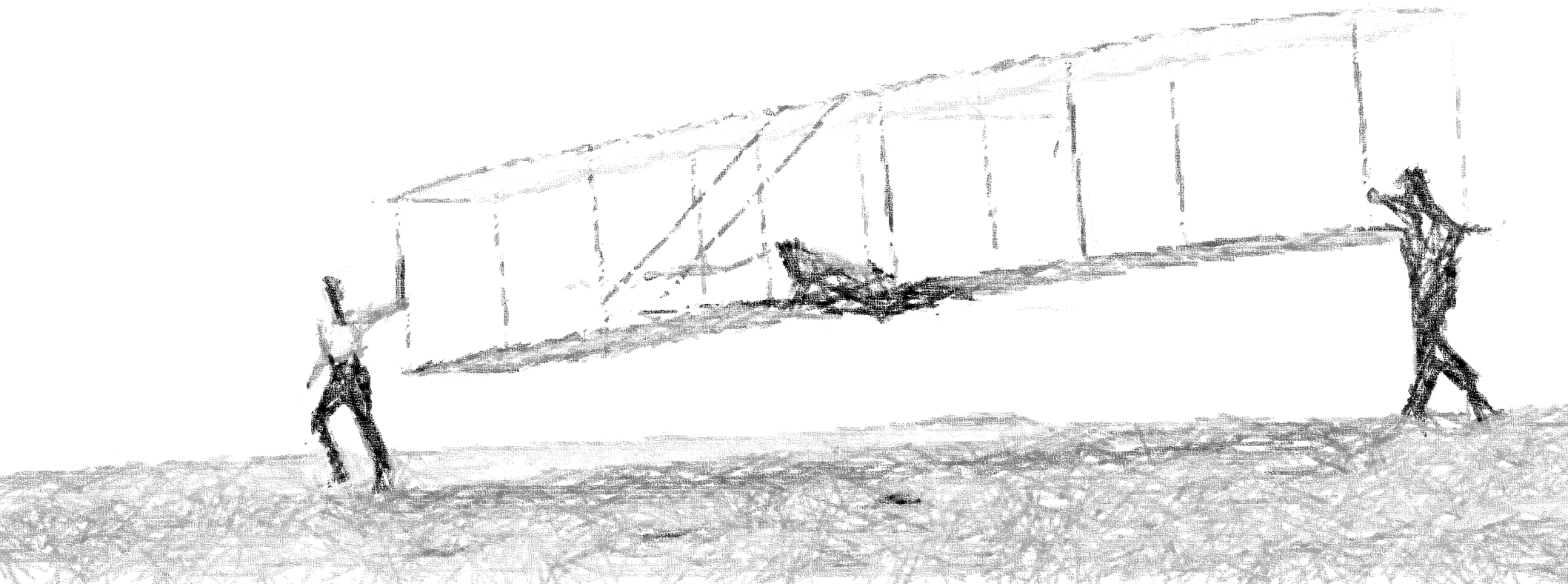




Mechanics applied to Aerospace Engineering

Course 251-14165 – BSc Aerospace Engineering

5. Relative motion



Contents

- Introduction to relative motion
- Position and translational motion of a reference frame
- Attitude of a reference frame
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- Angular motion of a reference frame
 - Angular velocity
 - Angular acceleration
- Coriolis theorem
- Relative kinematics of the point particle
 - Position, velocity, and acceleration formulas
- Relative dynamics of the point particle
 - Forces of inertia

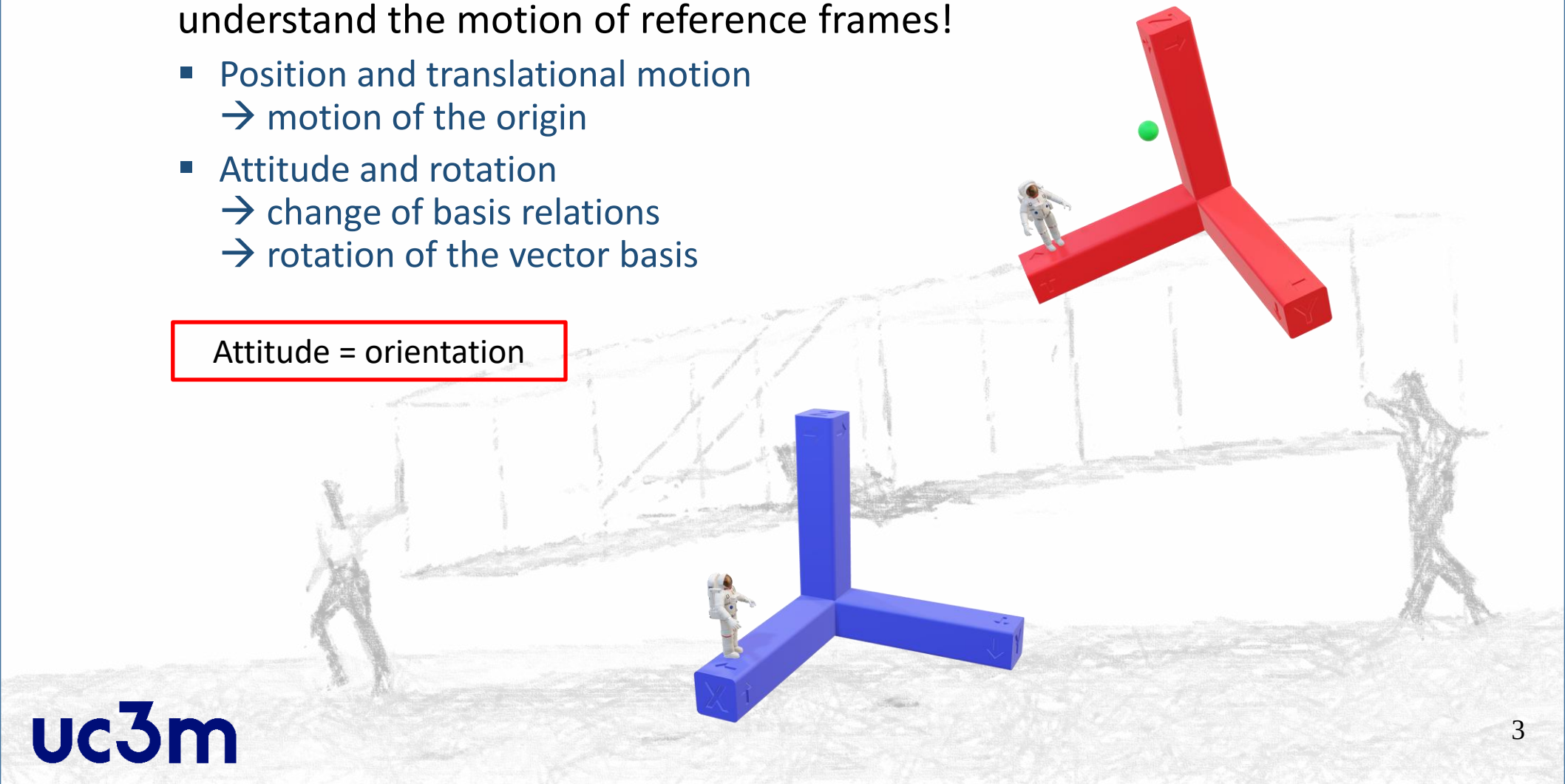
Relative motion

- Different observers describe the motion of a particle P differently

$$\mathbf{r}_0^P \neq \mathbf{r}_1^P; \quad \mathbf{v}_0^P \neq \mathbf{v}_1^P; \quad \mathbf{a}_0^P \neq \mathbf{a}_1^P$$

- An observer $\equiv$ a reference frame
- First step to understand relative motion:
understand the motion of reference frames!
 - Position and translational motion
→ motion of the origin
 - Attitude and rotation
→ change of basis relations
→ rotation of the vector basis

Attitude = orientation



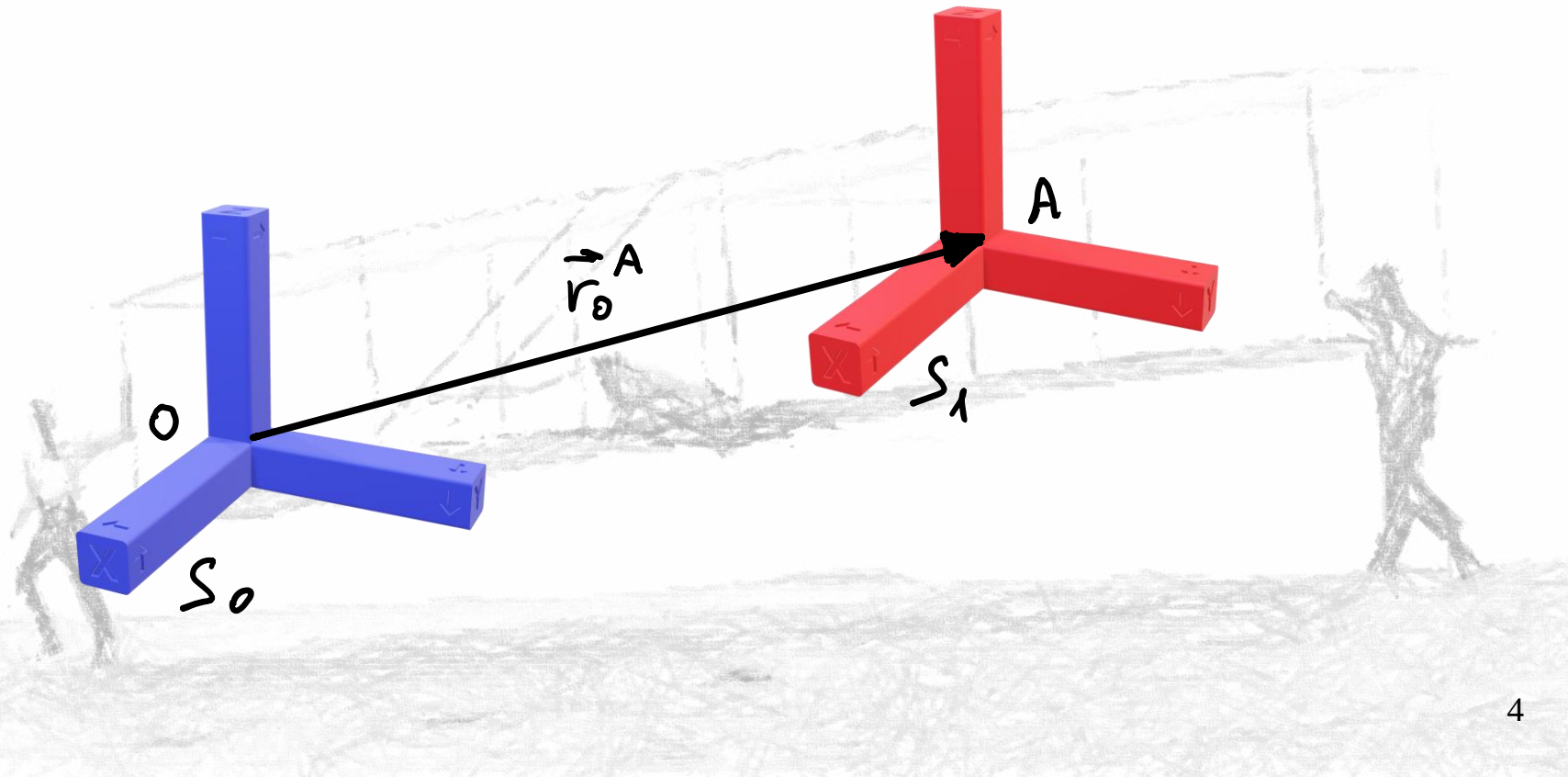
Position and translational motion

- Consider two reference frames, $S_0 = \{O; B_0\}$ and $S_1 = \{A; B_1\}$.
- The *translational* motion of S_1 with respect to S_0 concerns the motion of point A as seen from S_0

$$\mathbf{r}_0^A; \quad \mathbf{v}_0^A; \quad \mathbf{a}_0^A$$

- Alternatively, we can describe the motion of O from S_1 with $\mathbf{r}_1^O, \mathbf{v}_1^O, \mathbf{a}_1^O$.
Observe that:

$$\mathbf{r}_1^O = -\mathbf{r}_0^A, \quad \text{but: } \mathbf{v}_1^O \neq -\mathbf{v}_0^A; \quad \mathbf{a}_1^O \neq -\mathbf{a}_0^A \text{ in general!}$$



Attitude of a reference frame

- The *attitude* of a reference frame S_1 with respect to another S_0 is its instantaneous orientation in space. The position of the origin does not matter: only the direction of the vectors of the bases
- There are several ways to describe attitude. The simplest is to define the vectors of B_1 in terms of the vectors of B_0 :

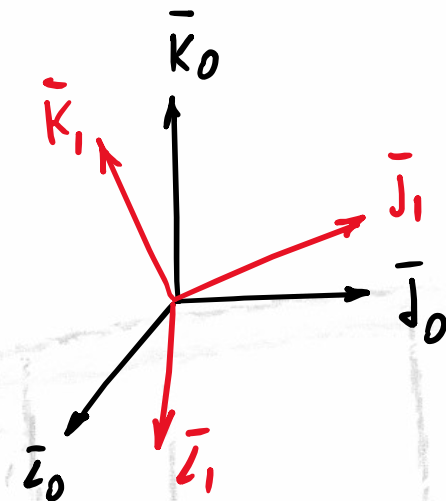
$$\begin{cases} \mathbf{i}_1 = \ell_{x1,x0}\mathbf{i}_0 + \ell_{x1,y0}\mathbf{j}_0 + \ell_{x1,z0}\mathbf{k}_0, \\ \mathbf{j}_1 = \ell_{y1,x0}\mathbf{i}_0 + \ell_{y1,y0}\mathbf{j}_0 + \ell_{y1,z0}\mathbf{k}_0, \\ \mathbf{k}_1 = \ell_{z1,x0}\mathbf{i}_0 + \ell_{z1,y0}\mathbf{j}_0 + \ell_{z1,z0}\mathbf{k}_0. \end{cases}$$

- Remember that any vector $\mathbf{b}$ can be written in either basis:

$$\begin{aligned} \mathbf{b} &= b_{x0}\mathbf{i}_0 + b_{y0}\mathbf{j}_0 + b_{z0}\mathbf{k}_0 \\ &= b_{x1}\mathbf{i}_1 + b_{y1}\mathbf{j}_1 + b_{z1}\mathbf{k}_1. \end{aligned}$$

Using the change of basis matrix:

$$\begin{bmatrix} b_{x0} \\ b_{y0} \\ b_{z0} \end{bmatrix}_0 = [{}_0R_1] \cdot \begin{bmatrix} b_{x1} \\ b_{y1} \\ b_{z1} \end{bmatrix}_1 \quad [{}_0R_1] = \begin{bmatrix} \ell_{x1,x0} & \ell_{y1,x0} & \ell_{z1,x0} \\ \ell_{x1,y0} & \ell_{y1,y0} & \ell_{z1,y0} \\ \ell_{x1,z0} & \ell_{y1,z0} & \ell_{z1,z0} \end{bmatrix}$$



Attitude of a reference frame

- The change of basis relations and matrix between two right-handed, orthonormal bases has several properties of interest:

- $[{}_0R_1]$ is an orthogonal transformation matrix

- The inverse transformation is equal to the transpose, $[{}_0R_1]^{-1} = [{}_0R_1]^T$

$$\begin{cases} \mathbf{i}_1 = \ell_{x1,x0}\mathbf{i}_0 + \ell_{x1,y0}\mathbf{j}_0 + \ell_{x1,z0}\mathbf{k}_0, \\ \mathbf{j}_1 = \ell_{y1,x0}\mathbf{i}_0 + \ell_{y1,y0}\mathbf{j}_0 + \ell_{y1,z0}\mathbf{k}_0, \\ \mathbf{k}_1 = \ell_{z1,x0}\mathbf{i}_0 + \ell_{z1,y0}\mathbf{j}_0 + \ell_{z1,z0}\mathbf{k}_0. \end{cases} \quad \begin{cases} \mathbf{i}_0 = \ell_{x0,x1}\mathbf{i}_1 + \ell_{x0,y1}\mathbf{j}_1 + \ell_{x0,z1}\mathbf{k}_1, \\ \mathbf{j}_0 = \ell_{y0,x1}\mathbf{i}_1 + \ell_{y0,y1}\mathbf{j}_1 + \ell_{y0,z1}\mathbf{k}_1, \\ \mathbf{k}_0 = \ell_{z0,x1}\mathbf{i}_1 + \ell_{z0,y1}\mathbf{j}_1 + \ell_{z0,z1}\mathbf{k}_1. \end{cases}$$

$$\begin{bmatrix} b_{x1} \\ b_{y1} \\ b_{z1} \end{bmatrix}_1 = [{}_1R_0] \cdot \begin{bmatrix} b_{x0} \\ b_{y0} \\ b_{z0} \end{bmatrix}_0 \quad \begin{bmatrix} b_{x0} \\ b_{y0} \\ b_{z0} \end{bmatrix}_0 = [{}_0R_1] \cdot \begin{bmatrix} b_{x1} \\ b_{y1} \\ b_{z1} \end{bmatrix}_1$$

- This means $\ell_{x0,x1} = \ell_{x1,x0}$, $\ell_{x0,y1} = \ell_{y1,x0}$, etc

- The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in B_0
- The rows of $[{}_0R_1]$ are the components of $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$ in B_1
- The eigenvalues of $[{}_0R_1]$ are $\lambda = 1, e^{i\beta}, e^{-i\beta}$.
- The transformation from B_1 to B_0 is a rotation:
 - The *rotation angle* about this axis of rotation is β
 - The *axis of rotation* is defined by the eigenspace associated to $\lambda = 1$. The components of any vector in this axis are identical in B_1 and B_0

Composition rules

- Imagine we have three bases: B_0, B_1, B_2 :

$$\begin{cases} \mathbf{i}_1 = \ell_{x1,x0}\mathbf{i}_0 + \ell_{x1,y0}\mathbf{j}_0 + \ell_{x1,z0}\mathbf{k}_0, \\ \mathbf{j}_1 = \ell_{y1,x0}\mathbf{i}_0 + \ell_{y1,y0}\mathbf{j}_0 + \ell_{y1,z0}\mathbf{k}_0, \\ \mathbf{k}_1 = \ell_{z1,x0}\mathbf{i}_0 + \ell_{z1,y0}\mathbf{j}_0 + \ell_{z1,z0}\mathbf{k}_0. \end{cases} \quad [{}_0R_1] = \begin{bmatrix} \ell_{x1,x0} & \ell_{y1,x0} & \ell_{z1,x0} \\ \ell_{x1,y0} & \ell_{y1,y0} & \ell_{z1,y0} \\ \ell_{x1,z0} & \ell_{y1,z0} & \ell_{z1,z0} \end{bmatrix}$$

$$\begin{cases} \mathbf{i}_2 = \ell_{x2,x1}\mathbf{i}_1 + \ell_{x2,y1}\mathbf{j}_1 + \ell_{x2,z1}\mathbf{k}_1, \\ \mathbf{j}_2 = \ell_{y2,x1}\mathbf{i}_1 + \ell_{y2,y1}\mathbf{j}_1 + \ell_{y2,z1}\mathbf{k}_1, \\ \mathbf{k}_2 = \ell_{z2,x1}\mathbf{i}_1 + \ell_{z2,y1}\mathbf{j}_1 + \ell_{z2,z1}\mathbf{k}_1. \end{cases} \quad [{}_1R_2] = \begin{bmatrix} \ell_{x2,x1} & \ell_{y2,x1} & \ell_{z2,x1} \\ \ell_{x2,y1} & \ell_{y2,y1} & \ell_{z2,y1} \\ \ell_{x2,z1} & \ell_{y2,z1} & \ell_{z2,z1} \end{bmatrix}$$

- The change of basis relations can be composed to write $\mathbf{i}_2$ as a function of $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$. For example →

$$\begin{aligned} \mathbf{i}_2 &= \ell_{x2,x1}(\ell_{x1,x0}\mathbf{i}_0 + \ell_{x1,y0}\mathbf{j}_0 + \ell_{x1,z0}\mathbf{k}_0) \\ &\quad + \ell_{x2,y1}(\ell_{y1,x0}\mathbf{i}_0 + \ell_{y1,y0}\mathbf{j}_0 + \ell_{y1,z0}\mathbf{k}_0) \\ &\quad + \ell_{x2,z1}(\ell_{z1,x0}\mathbf{i}_0 + \ell_{z1,y0}\mathbf{j}_0 + \ell_{z1,z0}\mathbf{k}_0), \end{aligned}$$

- Equivalently, matrices can be composed:

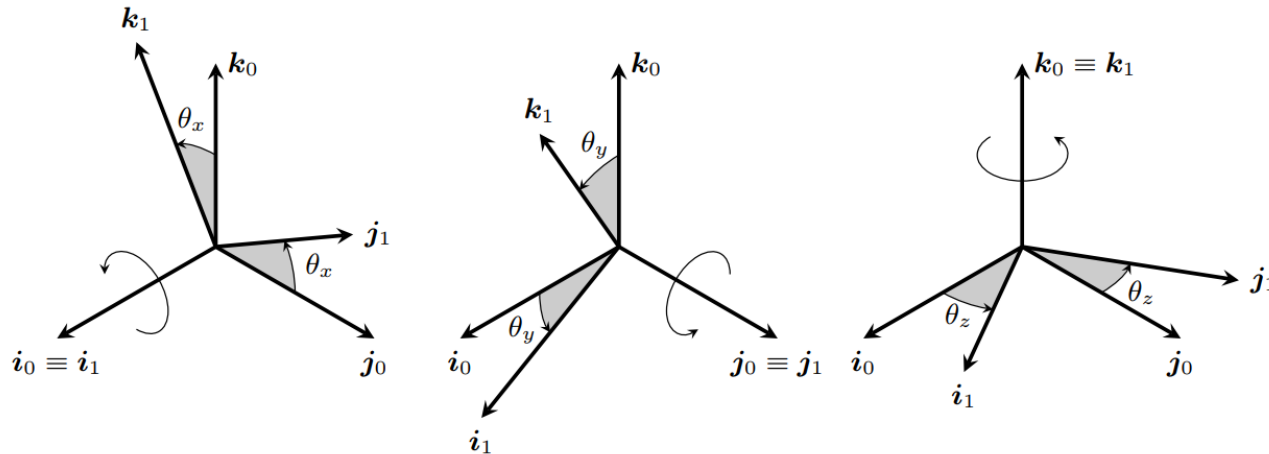
$$\begin{bmatrix} b_{x0} \\ b_{y0} \\ b_{z0} \end{bmatrix} = [{}_0R_1] \cdot \begin{bmatrix} b_{x1} \\ b_{y1} \\ b_{z1} \end{bmatrix} \quad \begin{bmatrix} b_{x0} \\ b_{y0} \\ b_{z0} \end{bmatrix} = [{}_0R_1] \cdot [{}_1R_2] \cdot \begin{bmatrix} b_{x2} \\ b_{y2} \\ b_{z2} \end{bmatrix}$$

$$\begin{bmatrix} b_{x1} \\ b_{y1} \\ b_{z1} \end{bmatrix} = [{}_1R_2] \cdot \begin{bmatrix} b_{x2} \\ b_{y2} \\ b_{z2} \end{bmatrix}$$

$$[{}_0R_2] = [{}_0R_1] \cdot [{}_1R_2]$$

Simple rotations

- We call *simple rotations* those whose axis of rotation is along one of the vectors of the basis



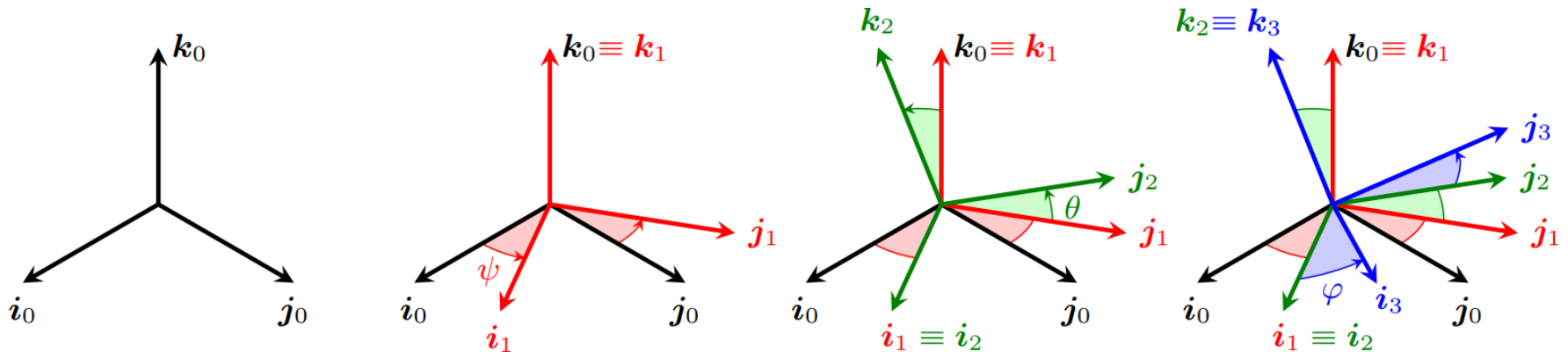
$$\begin{cases} i_1 = i_0, \\ j_1 = \cos \theta_x j_0 + \sin \theta_x k_0, \\ k_1 = -\sin \theta_x j_0 + \cos \theta_x k_0, \end{cases} \quad [{}_0R_1(x, \theta_x)] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_x & -\sin \theta_x \\ 0 & \sin \theta_x & \cos \theta_x \end{bmatrix}$$

$$\begin{cases} i_1 = \cos \theta_y i_0 - \sin \theta_y k_0, \\ j_1 = j_0, \\ k_1 = \sin \theta_y i_0 + \cos \theta_y k_0, \end{cases} \quad [{}_0R_1(y, \theta_y)] = \begin{bmatrix} \cos \theta_y & 0 & \sin \theta_y \\ 0 & 1 & 0 \\ -\sin \theta_y & 0 & \cos \theta_y \end{bmatrix}$$

$$\begin{cases} i_1 = \cos \theta_z i_0 + \sin \theta_z j_0, \\ j_1 = -\sin \theta_z i_0 + \cos \theta_z j_0, \\ k_1 = k_0. \end{cases} \quad [{}_0R_1(z, \theta_z)] = \begin{bmatrix} \cos \theta_z & -\sin \theta_z & 0 \\ \sin \theta_z & \cos \theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Euler angles

- It turns out it is possible to represent any attitude as the composition of three simple rotations about different axes. For example, to relate B_0 to B_3 , introducing the auxiliary bases B_1 and B_2 :

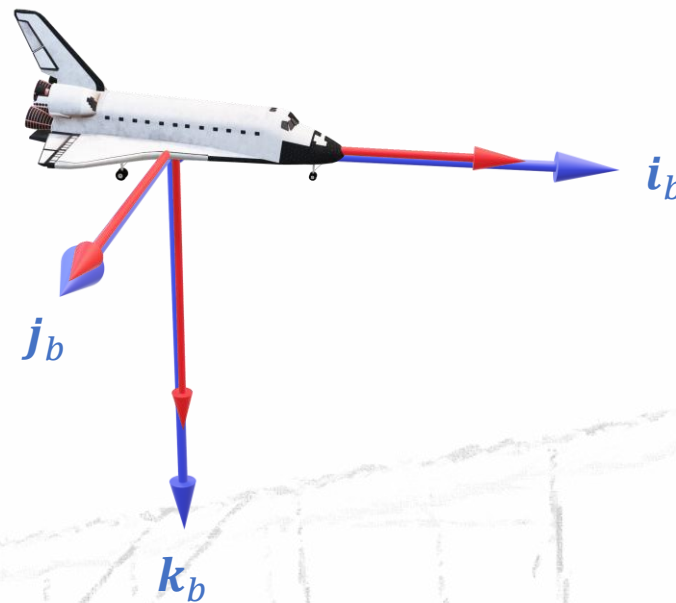


$$[{}_0R_3] = [{}_0R_1] \cdot [{}_1R_2] \cdot [{}_2R_3]$$

- The angles ψ (precession), θ (nutation), ϕ (spin) are known as *standard Euler angles*, and can be used to define the attitude
 - They are also known as 3-1-3 Euler angles (after the order of rotation: third axis, then first axis, then third again).
- There are many other possible choices to define alternative Euler angles. Another common choice is 3-2-1 (yaw, pitch, roll)
 - The only rule is: you cannot repeat the same axis sequentially

Example

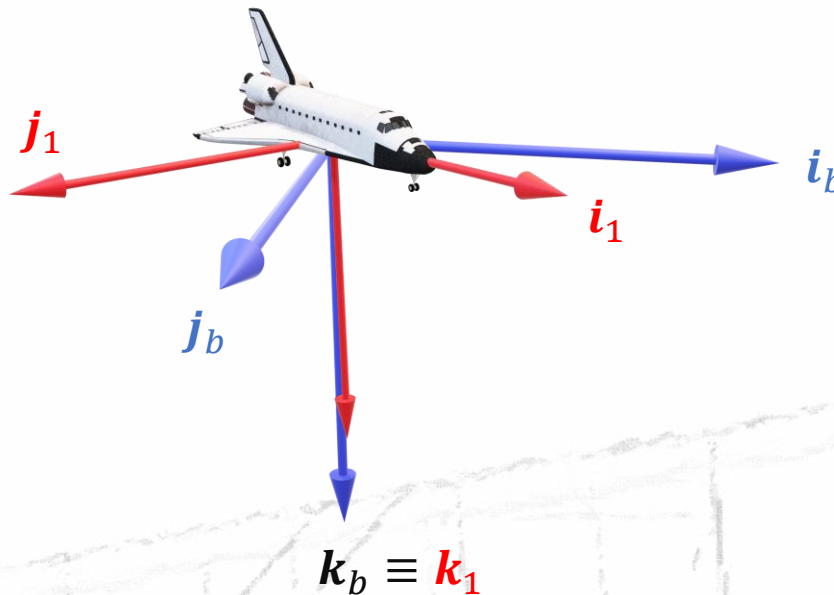
- We want to describe the attitude of the red reference frame S_r (attached to the space shuttle for visualization) with respect to the blue reference frame S_b with 3-2-1 Euler angles (conventional in flight mechanics)
 - Begin with both frames coincident



Example

- We want to describe the attitude of the red reference frame S_r (attached to the space shuttle for visualization) with respect to the blue reference frame S_b with 3-2-1 Euler angles (conventional in flight mechanics)

- Begin with both frames coincident
- Perform first rotation (about k_b)

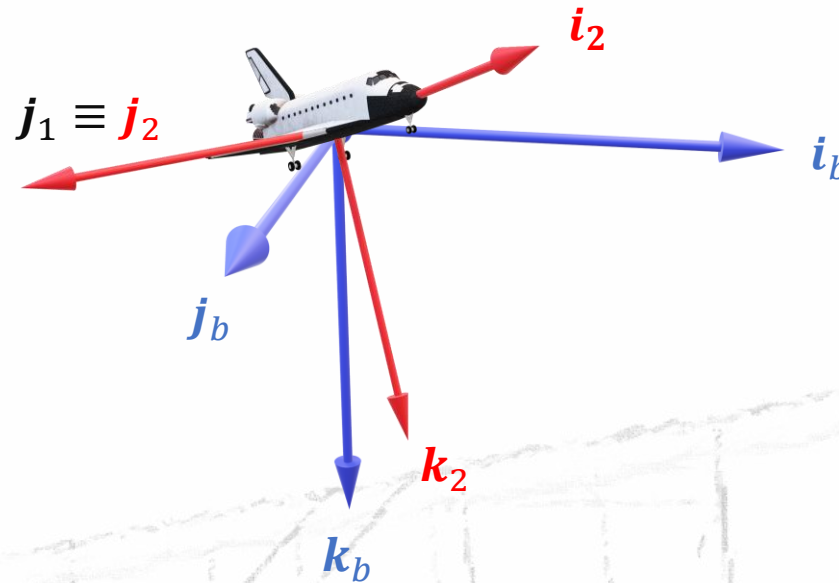


$$\begin{cases} i_1 = \cos \psi i_b + \sin \psi j_b, \\ j_1 = -\sin \psi i_b + \cos \psi j_b, \\ k_1 = k_b. \end{cases}$$

Example

- We want to describe the attitude of the red reference frame S_r (attached to the space shuttle for visualization) with respect to the blue reference frame S_b with 3-2-1 Euler angles (conventional in flight mechanics)

- Begin with both frames coincident
- Perform first rotation (about $\mathbf{k}_b$)
- Perform second rotation (about $\mathbf{j}_1$)

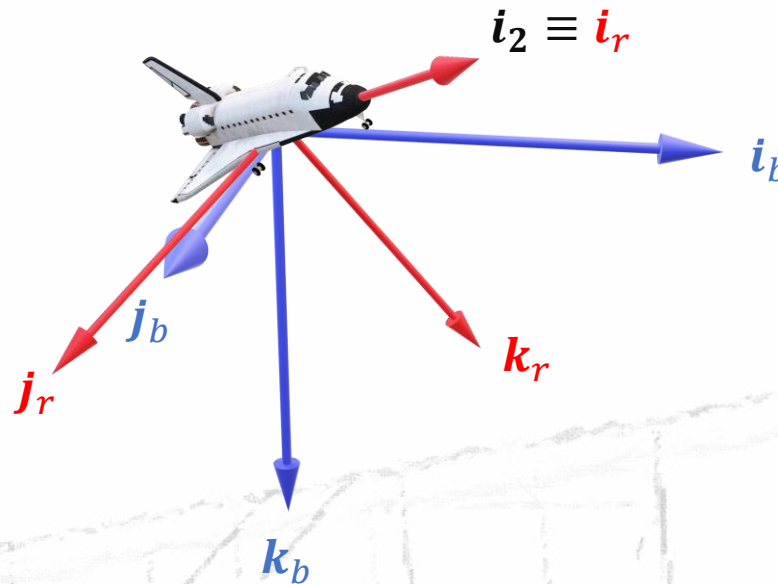


$$\begin{cases} i_2 = \cos \theta i_1 - \sin \theta k_1, \\ j_2 = j_1, \\ k_2 = \sin \theta i_1 + \cos \theta k_1, \end{cases}$$

Example

- We want to describe the attitude of the red reference frame S_r (attached to the space shuttle for visualization) with respect to the blue reference frame S_b with 3-2-1 Euler angles (conventional in flight mechanics)

- Begin with both frames coincident
- Perform first rotation (about $\mathbf{k}_b$)
- Perform second rotation (about $\mathbf{j}_1$)
- Perform second rotation (about $\mathbf{i}_2$)



$$\begin{cases} \mathbf{i}_r = \mathbf{i}_2, \\ \mathbf{j}_r = \cos \varphi \mathbf{j}_2 + \sin \varphi \mathbf{k}_2, \\ \mathbf{k}_r = -\sin \varphi \mathbf{j}_2 + \cos \varphi \mathbf{k}_2, \end{cases}$$

Angular motion of a reference frame

- So far we have discussed stationary changes of basis. But vector bases can rotate with respect to each other, as a function of time
- The rotation of B_1 with respect to B_0 in time can be represented by its *instantaneous angular velocity vector* $\boldsymbol{\omega}_{10}$
 - Its direction is the direction of the instantaneous axis of rotation of B_1
 - Its magnitude is the angular rate
- The vectors of basis B_1 change in time when observed from S_0 as:

Note that $\boldsymbol{\omega}_{10}$ and the motion of the origin of S_1 are unrelated: two completely different things!

$$\left. \frac{d\mathbf{i}_1}{dt} \right|_0 = \boldsymbol{\omega}_{10} \times \mathbf{i}_1, \quad \left. \frac{d\mathbf{j}_1}{dt} \right|_0 = \boldsymbol{\omega}_{10} \times \mathbf{j}_1, \quad \left. \frac{d\mathbf{k}_1}{dt} \right|_0 = \boldsymbol{\omega}_{10} \times \mathbf{k}_1$$

- Any constant vector $\mathbf{c} = c_{x1}\mathbf{i}_1 + c_{y1}\mathbf{j}_1 + c_{z1}\mathbf{k}_1$ in S_1 also changes in time like that in S_0 :

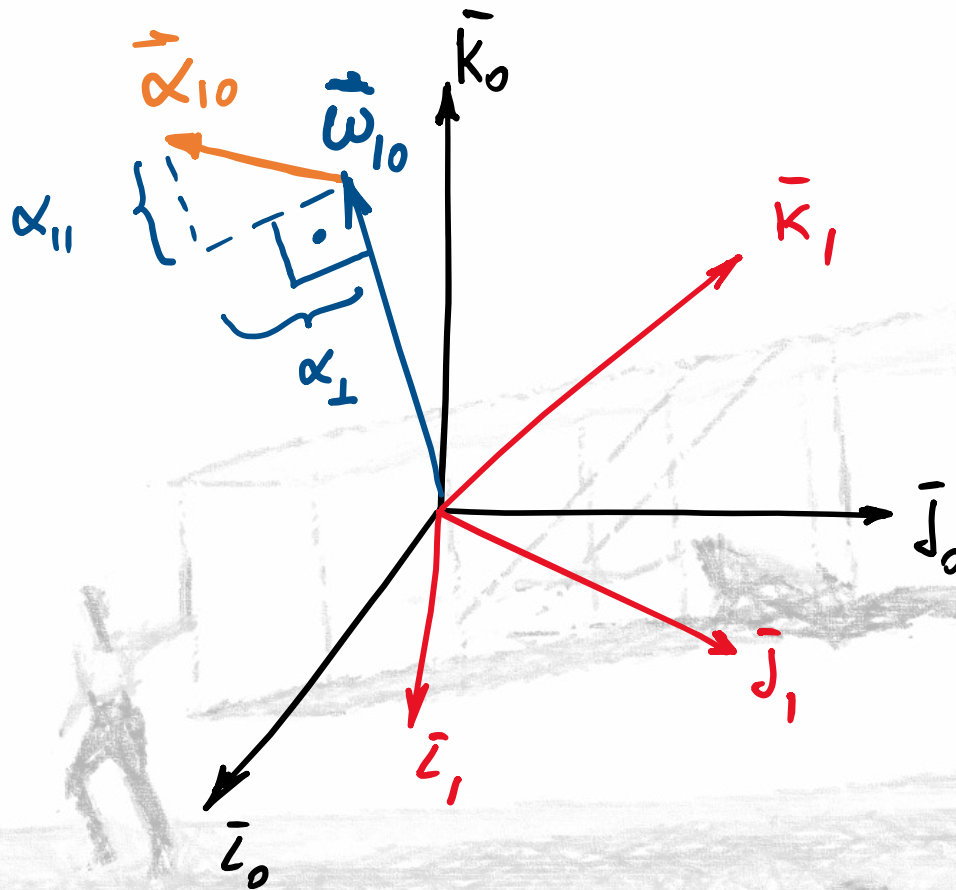
$$\begin{aligned} \left. \frac{d\mathbf{c}}{dt} \right|_0 &= c_{x1} \left. \frac{d\mathbf{i}_1}{dt} \right|_0 + c_{y1} \left. \frac{d\mathbf{j}_1}{dt} \right|_0 + c_{z1} \left. \frac{d\mathbf{k}_1}{dt} \right|_0 \\ &= c_{x1} \boldsymbol{\omega}_{10} \times \mathbf{i}_1 + c_{y1} \boldsymbol{\omega}_{10} \times \mathbf{j}_1 + c_{z1} \boldsymbol{\omega}_{10} \times \mathbf{k}_1 \\ &= \boldsymbol{\omega}_{10} \times c_{x1} \mathbf{i}_1 + \boldsymbol{\omega}_{10} \times c_{y1} \mathbf{j}_1 + \boldsymbol{\omega}_{10} \times c_{z1} \mathbf{k}_1 = \boldsymbol{\omega}_{10} \times \mathbf{c} \end{aligned}$$

Angular motion of a reference frame

- Moreover, the rotation can be accelerated. We define the *instantaneous angular acceleration vector* α_{10} as:

$$\alpha_{10} = \left. \frac{d\omega_{10}}{dt} \right|_0$$

- Note that the direction of α_{10} may be different from the direction of ω_{10}



Note that ω_{10} does not describe the current attitude of S_1 , only how it is changing in time

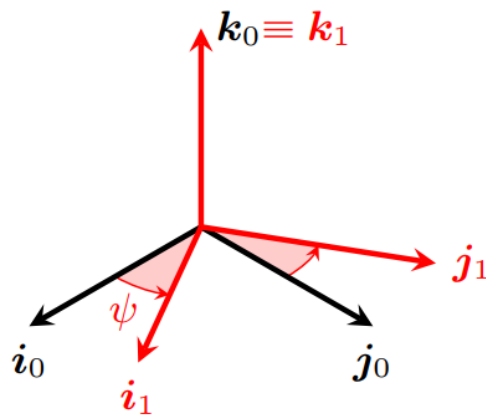
The parallel component of α_{10} describes how the magnitude of ω_{10} changes in time

The perpendicular component of α_{10} describes how the direction of ω_{10} changes in time

Example

- At time $t = 0$, reference frames S_1 and S_0 coincide. Thereafter, S_1 performs a simple rotation about the $\mathbf{k}_0$ axis with an angle $\psi = \psi(t)$

$$\begin{cases} \mathbf{i}_1 = \cos \psi \mathbf{i}_0 + \sin \psi \mathbf{j}_0, \\ \mathbf{j}_1 = -\sin \psi \mathbf{i}_0 + \cos \psi \mathbf{j}_0, \\ \mathbf{k}_1 = \mathbf{k}_0. \end{cases}$$



$$\left. \frac{d\mathbf{i}_1}{dt} \right|_0 = \boldsymbol{\omega}_{10} \times \mathbf{i}_1$$

$$\left. \frac{d\mathbf{j}_1}{dt} \right|_0 = \boldsymbol{\omega}_{10} \times \mathbf{j}_1$$

$$\left. \frac{d\mathbf{k}_1}{dt} \right|_0 = \boldsymbol{\omega}_{10} \times \mathbf{k}_1$$

Coriolis theorem

- Consider a vector $\mathbf{b} = b_{x1}(t)\mathbf{i}_1 + b_{y1}(t)\mathbf{j}_1 + b_{z1}(t)\mathbf{k}_1$ that is not constant in S_1 .

- Derivative in S_1 :

$$\left. \frac{d\mathbf{b}}{dt} \right|_1 = \frac{db_{x1}}{dt} \mathbf{i}_1 + \frac{db_{y1}}{dt} \mathbf{j}_1 + \frac{db_{z1}}{dt} \mathbf{k}_1$$

- Derivative in S_0 :

$$\left. \frac{d\mathbf{b}}{dt} \right|_0 = \frac{db_{x1}}{dt} \mathbf{i}_1 + \frac{db_{y1}}{dt} \mathbf{j}_1 + \frac{db_{z1}}{dt} \mathbf{k}_1 + b_{x1} \left. \frac{d\mathbf{i}_1}{dt} \right|_0 + b_{y1} \left. \frac{d\mathbf{j}_1}{dt} \right|_0 + b_{z1} \left. \frac{d\mathbf{k}_1}{dt} \right|_0$$

- Therefore:

$$\left. \frac{d\mathbf{b}}{dt} \right|_0 = \left. \frac{d\mathbf{b}}{dt} \right|_1 + \boldsymbol{\omega}_{10} \times \mathbf{b}$$

Coriolis
Theorem

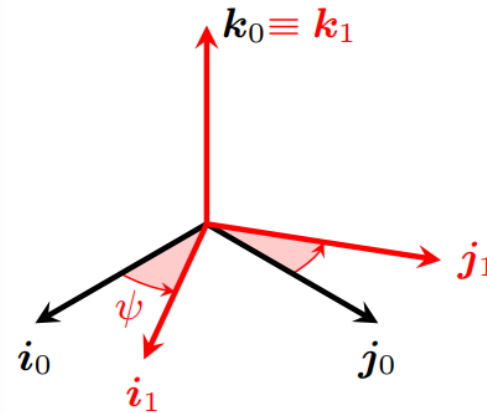
- The derivative of vector $\mathbf{b}$ in S_0 equals the derivative of $\mathbf{b}$ in S_1 plus the rotation of $\mathbf{b}$ due to the angular motion of S_1 with respect to S_0

Remember: a vector can be defined in one reference frame, be differentiated in another, and be expressed in yet another

Example

➤ Consider $\mathbf{b} = t\mathbf{i}_1 + t^2\mathbf{j}_1 + t^3\mathbf{k}_1$. Compute its derivative in S_1 and S_0

$$\begin{cases} \mathbf{i}_1 = \cos \psi \mathbf{i}_0 + \sin \psi \mathbf{j}_0, \\ \mathbf{j}_1 = -\sin \psi \mathbf{i}_0 + \cos \psi \mathbf{j}_0, \\ \mathbf{k}_1 = \mathbf{k}_0. \end{cases}$$



Composition of angular motions

- Consider S_2 rotating with respect to S_1 with $\boldsymbol{\omega}_{21}$, and S_1 rotating with respect to S_0 with $\boldsymbol{\omega}_{10}$. The rotation of S_2 with respect to S_0 is given by:

$$\boldsymbol{\omega}_{20} = \boldsymbol{\omega}_{21} + \boldsymbol{\omega}_{10}$$

- The rule of composition of angular accelerations is more complex. Applying Coriolis theorem above:

$$\left. \frac{d\boldsymbol{\omega}_{20}}{dt} \right|_0 = \left. \frac{d\boldsymbol{\omega}_{21}}{dt} \right|_0 + \left. \frac{d\boldsymbol{\omega}_{10}}{dt} \right|_0$$

$$\left. \frac{d\boldsymbol{\omega}_{21}}{dt} \right|_1 + \boldsymbol{\omega}_{10} \times \boldsymbol{\omega}_{21}$$

$$\boldsymbol{\alpha}_{20} = \boldsymbol{\alpha}_{21} + \boldsymbol{\omega}_{10} \times \boldsymbol{\omega}_{21} + \boldsymbol{\alpha}_{10}$$

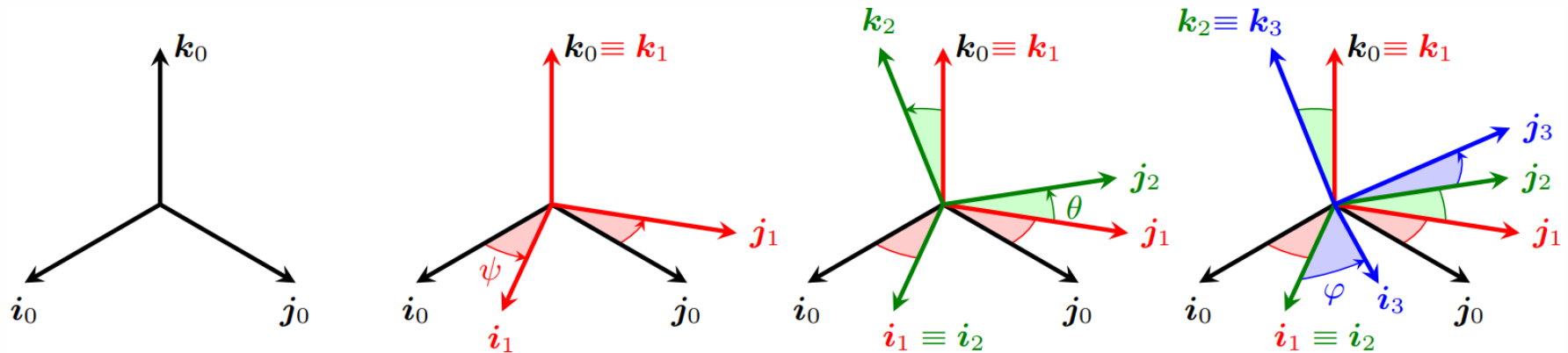
- The rotation of S_0 with respect to S_1 satisfies:

$$\boldsymbol{\omega}_{01} = -\boldsymbol{\omega}_{10}; \quad \boldsymbol{\alpha}_{01} = -\boldsymbol{\alpha}_{10}$$

Angular velocity from simple rotations

- Consider the three sequential simple rotations associated to 3-1-3 Euler angles as a function of time, $\psi(t), \theta(t), \varphi(t)$:

$$S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow S_3$$



- What is the angular velocity of each simple rotation? and for the combined rotation?

$$\omega_{10} = \dot{\psi} k_0 \equiv \dot{\psi} k_1$$

$$\omega_{21} = \dot{\theta} i_1 \equiv \dot{\theta} i_2$$

$$\omega_{32} = \dot{\varphi} k_2 \equiv \dot{\varphi} k_3$$

$$\omega_{30} = \omega_{32} + \omega_{21} + \omega_{10} = \dot{\varphi} k_2 + \dot{\theta} i_1 + \dot{\psi} k_0$$

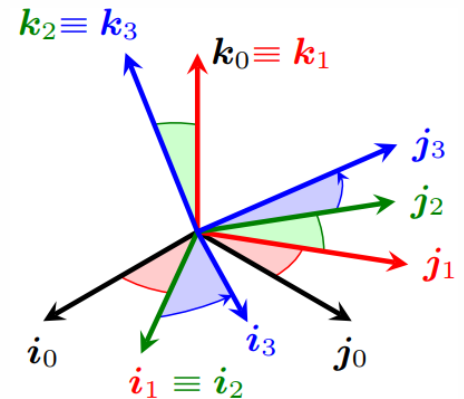
Remember: a vector can be defined in one reference frame, be differentiated in another, and be expressed in a different vector basis (or, in general: as linear combination of other vectors)

Angular velocity from simple rotations

$$\omega_{30} = \omega_{32} + \omega_{21} + \omega_{10} = \dot{\varphi} \mathbf{k}_2 + \dot{\theta} \mathbf{i}_1 + \dot{\psi} \mathbf{k}_0$$

- Transforming everything to B_0 :

$$\begin{aligned} \omega_{30} = & (\dot{\theta} \cos \psi + \dot{\varphi} \sin \psi \sin \theta) \mathbf{i}_0 \\ & + (\dot{\theta} \sin \psi - \dot{\varphi} \cos \psi \sin \theta) \mathbf{j}_0 \\ & + (\dot{\varphi} \cos \theta + \dot{\psi}) \mathbf{k}_0. \end{aligned}$$



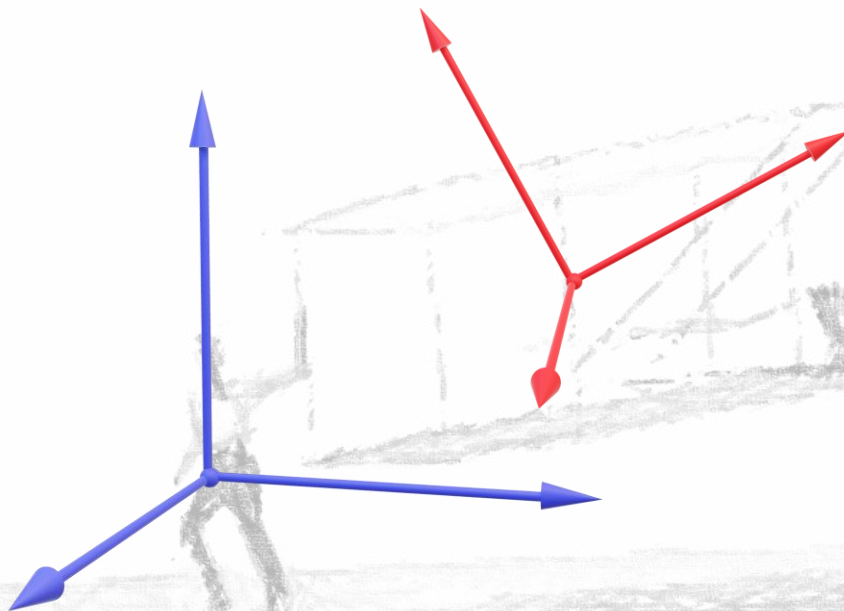
- Transforming everything to B_3

$$\begin{aligned} \omega_{30} = & (\dot{\theta} \cos \varphi + \dot{\psi} \sin \varphi \sin \theta) \mathbf{i}_3 \\ & + (-\dot{\theta} \sin \varphi + \dot{\psi} \cos \varphi \sin \theta) \mathbf{j}_3 \\ & + (\dot{\varphi} + \dot{\psi} \cos \theta) \mathbf{k}_3. \end{aligned}$$

Exercise: do the same for 3-2-1
(yaw-pitch-roll) Euler angles

Configuration and motion of a reference frame

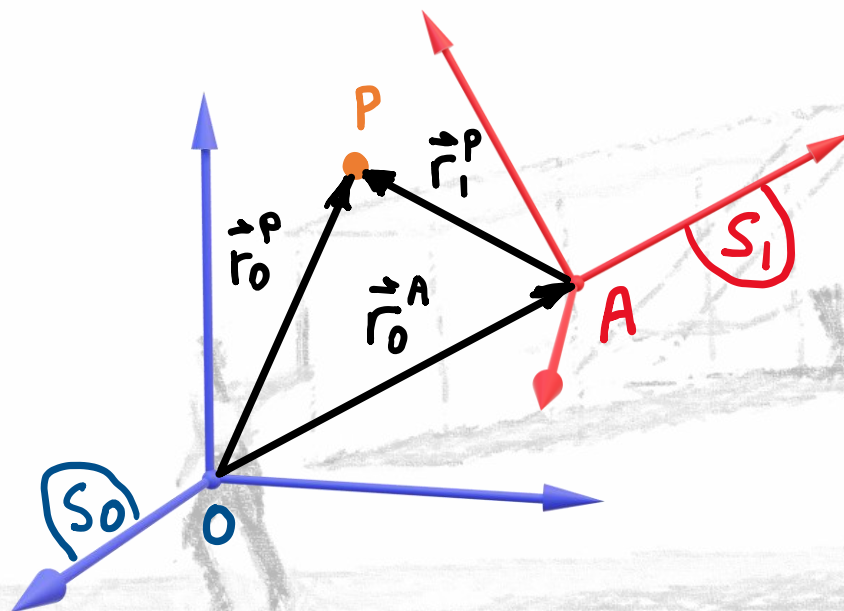
- The configuration and motion of $S_1 = \{A; B_1\}$ with respect to S_0 is completely specified if we provide:
 - The position of its origin A as a function of time: $\mathbf{r}_0^A(t)$
 - The attitude of its basis B_1 as a function of time: $\mathbf{i}_1(t), \mathbf{j}_1(t), \mathbf{k}_1(t)$
 - This can be done e.g. in terms of Euler angles $\psi(t), \theta(t), \varphi(t)$
- From this information we can compute:
 - The velocity and acceleration of the origin A : $\mathbf{v}_0^A(t), \mathbf{a}_0^A(t)$
 - The angular velocity and acceleration of the basis B_1 : $\boldsymbol{\omega}_{10}(t), \boldsymbol{\alpha}_{10}(t)$



The expressions of relative motion are particularly useful when the motion of S_1 is already known a priori. In this case, none of these functions of time are unknowns

Relative kinematics

- Quite often, we have more than one observer, and we need to relate the position, velocity, and acceleration of P that each one sees
 - E.g. We have sensors on an experimental aircraft to take measurements of a hurricane. We need to take into account the motion of the aircraft to “translate” those measurements to a ground-based reference frame
- In other cases, it may be simpler to obtain these vectors in a moving reference frame than in the fixed reference frame
- The relation between the position vectors in S_1 and S_0 can be obtained with a simple vector sum:



$$\vec{OP} = \vec{OA} + \vec{AP}$$

$$\vec{r}_0^P = \vec{r}_0^A + \vec{r}_1^P$$

- The “absolute position $\vec{r}_0^P$ ” (in S_0) equals the sum of the position of the origin of S_1 , $\vec{r}_0^A$ plus the “relative position $\vec{r}_1^P$ ” (in S_1)

Note that calling a reference frame “fixed” or “moving” and a position “absolute” or “relative” is completely arbitrary and subjective

Relative kinematics

$$\mathbf{r}_0^P = \mathbf{r}_0^A + \mathbf{r}_1^P$$

- To obtain the connection between velocity vectors in S_1 and S_0 we take derivatives:

$$\left. \frac{d\mathbf{r}_0^P}{dt} \right|_0 = \left. \frac{d\mathbf{r}_0^A}{dt} \right|_0 + \left. \frac{d\mathbf{r}_1^P}{dt} \right|_0$$

$$\mathbf{v}_0^P = \mathbf{v}_0^A + \left. \frac{d\mathbf{r}_1^P}{dt} \right|_1 + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P$$

$$\mathbf{v}_1^P$$

$$\mathbf{v}_0^P = \mathbf{v}_0^A + \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P$$

- The “absolute velocity $\mathbf{v}_0^P$ ” (in S_0) equals the sum of the velocity of the origin of S_1 , $\mathbf{v}_0^A$ plus the “relative velocity $\mathbf{v}_1^P$ ” (in S_1), plus the term $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P$ associated to the rotation of S_1 with respect to S_0

Relative kinematics

$$\mathbf{v}_0^P = \mathbf{v}_0^A + \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P$$

- To obtain the connection between acceleration vectors in S_1 and S_0 we take derivatives a second time:

$$\begin{aligned} \mathbf{a}_0^P \quad \frac{d\mathbf{v}_0^P}{dt} \Big|_0 &= \frac{d\mathbf{v}_0^A}{dt} \Big|_0 + \frac{d\mathbf{v}_1^P}{dt} \Big|_0 + \frac{d}{dt} \Big|_0 (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) \\ &= \mathbf{a}_0^A + \frac{d\mathbf{v}_1^P}{dt} \Big|_1 + \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P + \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times \frac{d\mathbf{r}_1^P}{dt} \Big|_0 \\ &= \mathbf{a}_0^A + \mathbf{a}_1^P + \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \frac{d\mathbf{r}_1^P}{dt} \Big|_1 + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P \\ \mathbf{a}_0^P &= \mathbf{a}_0^A + \mathbf{a}_1^P + \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P \end{aligned}$$

- The “absolute acceleration $\mathbf{a}_0^P$ ” (in S_0) equals the sum of the acceleration of the origin of S_1 , $\mathbf{a}_0^A$ plus the “relative acceleration $\mathbf{a}_1^P$ ” (in S_1), plus three terms associated to the rotation of S_1 with respect to S_0 : Euler, centripetal, and Coriolis accelerations

Relative kinematics

➤ In summary:

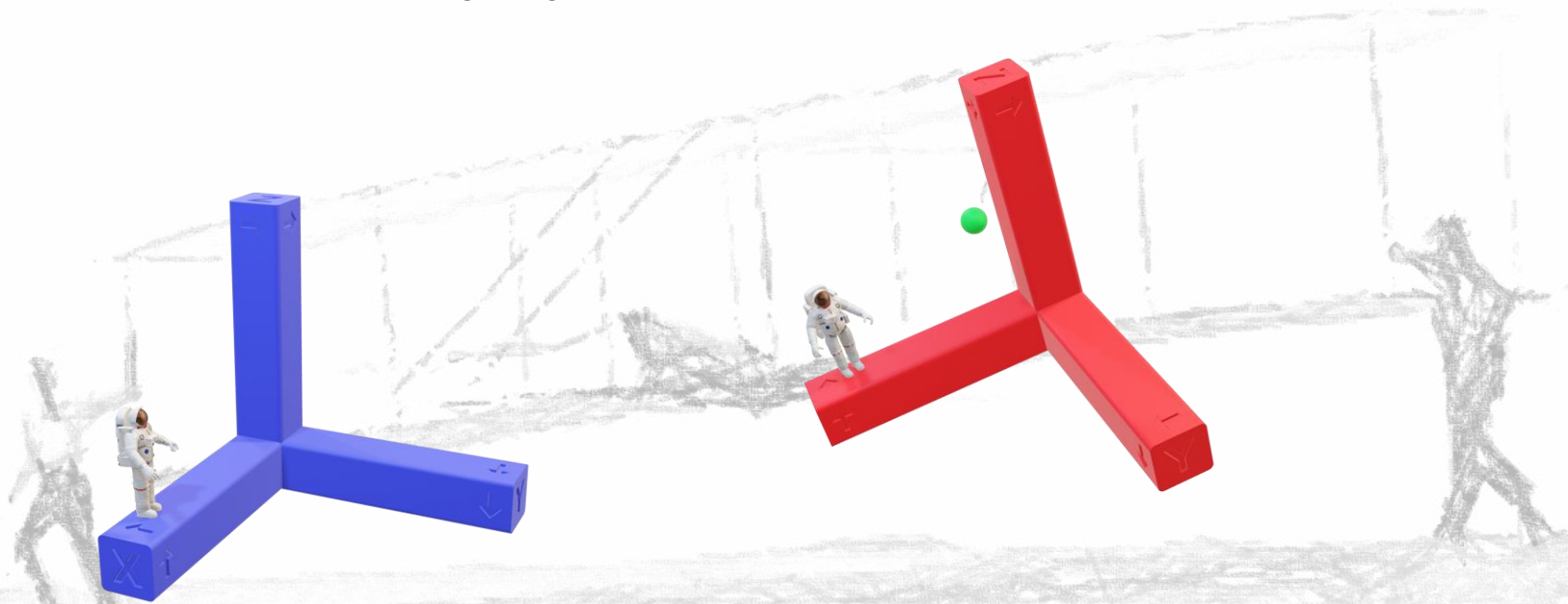
$$\mathbf{r}_0^P = \mathbf{r}_0^A + \mathbf{r}_1^P$$

$$\mathbf{v}_0^P = \mathbf{v}_0^A + \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P$$

$$\mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_1^P + \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$$

- Note that the right-hand sides can be computed using relative quantities only $(\mathbf{r}_1^P, \mathbf{v}_1^P, \mathbf{a}_1^P)$ plus the information on the motion of S_1 :

$$\mathbf{r}_0^A, \mathbf{v}_0^A, \mathbf{a}_0^A, \boldsymbol{\omega}_{10}, \boldsymbol{\alpha}_{10}$$



Relative dynamics

- Newton's laws of motion are only valid in an inertial reference frame
- Assuming S_0 is inertial, we can get the equations of motion of P by doing:

$$m^P \mathbf{a}_0^P = \mathbf{F}$$

- If S_1 is accelerating or rotating relative to S_0 , then it is non-inertial. Can we fix Newton's laws of motion so that we can apply them in S_1 ?
 - Consider the expression of the acceleration

$$\mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_1^P + \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$$

$$m^P [\mathbf{a}_0^A + \mathbf{a}_1^P + \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P] = \mathbf{F}$$

$$m^P \mathbf{a}_1^P = \mathbf{F} - m^P \mathbf{a}_0^A - m^P \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P - m^P \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) - 2m^P \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$$

$$m^P \mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$$

- Newton's laws can be applied to a non-inertial reference frame if we introduce a *fictitious* or *inertia force*:

$$\mathbf{F}_i = -m^P \mathbf{a}_0^A - m^P \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P - m^P \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) - 2m^P \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$$

Inertia forces

- To get the equation of motion of P in a non-inertial reference frame we must add the *inertia forces* to any *actual forces* in the problem:

$$\mathbf{F}_i = -m^P \mathbf{a}_0^A - m^P \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P - m^P \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) - 2m^P \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$$

- $-m^P \mathbf{a}_0^A$ is the force due to the acceleration of the origin
 - $-m^P \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P$ is Euler's force
 - $-m^P \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P)$ is the centrifugal force due to the rotation of S_1
 - $-2m^P \boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$ is the Coriolis force
- All forces of inertia are proportional to the mass of the particle
- In fact, force terms 1 to 3 are locally indistinguishable from a gravitational force field
- An observer in S_1 cannot tell them apart from an actual gravitational force
- Force 4 (Coriolis force) is a bit different: it is zero if the particle is at rest, and only makes its appearance when P has some relative velocity perpendicular to $\boldsymbol{\omega}_{10}$

We can now obtain the equations of motion of a particle in inertial and non-inertial reference frames. Try to choose the simplest option for each problem

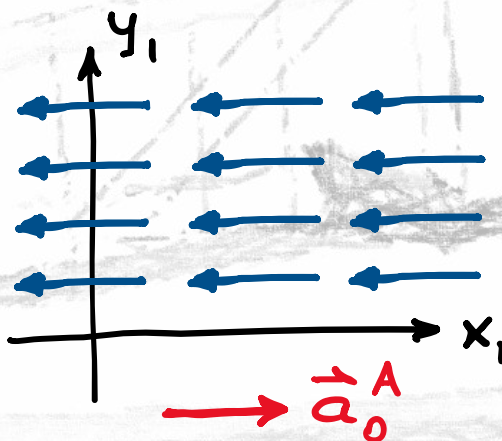
Inertia forces

We can always define the kinetic energy T_1^P and the mechanical energy E_1^P of P in S_1 even if it is non-inertial. If the necessary conditions are met (same as before), it is a conserved quantity of motion. Similarly, we can define the angular momentum $\mathbf{H}_{A1}^P$, or any other quantity we have used so far

- The inertial force due to the acceleration of the origin has a potential energy form in S_1 if $\mathbf{a}_0^A$ is constant:

$$V_1 = m^P \mathbf{a}_0^A \cdot \mathbf{r}_1^P; \quad \mathbf{F}_A = -\nabla V_1 = -m^P \mathbf{a}_0^A$$

- We can think of $\mathbf{F}_A$ as a homogeneous force field in S_1 :



Inertia forces

- The centrifugal force can be rewritten as follows. Let

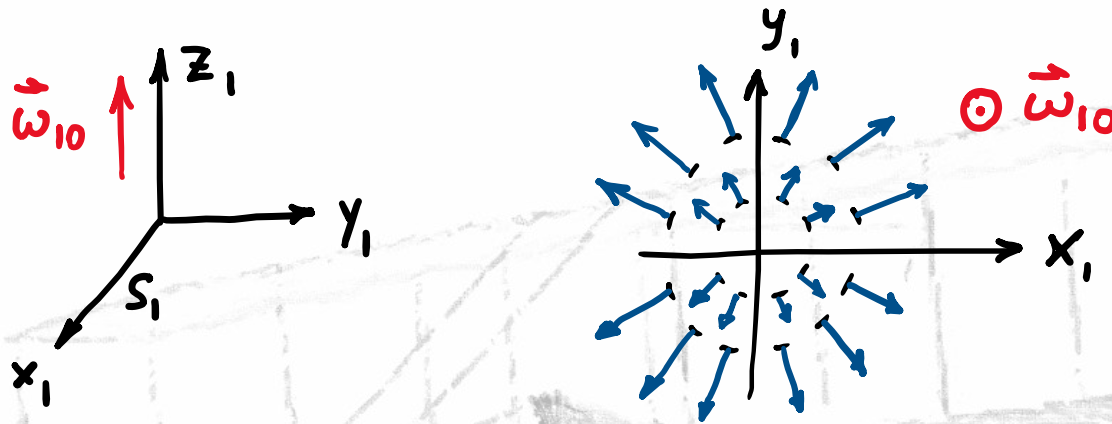
$$\mathbf{r}_1^P = \mathbf{r}_{\parallel} + \mathbf{r}_{\perp}; \quad \boldsymbol{\omega}_{10} = \omega_{10} \mathbf{e}_{\omega}$$

$$\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) = (\boldsymbol{\omega}_{10} \cdot \mathbf{r}_1^P) \boldsymbol{\omega}_{10} - (\omega_{10})^2 \mathbf{r}_1^P$$

$$(\omega_{10})^2 [r_{\parallel} \mathbf{e}_{\omega} - \mathbf{r}_1^P] \quad -(\omega_{10})^2 \mathbf{r}_{\perp}$$

- $\mathbf{F}_C = -m^P \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) = m^P (\omega_{10})^2 \mathbf{r}_{\perp}$

- $\mathbf{F}_C$ is proportional to $(\omega_{10})^2$ and $r_{\perp}$, and points in the direction of $\mathbf{r}_{\perp}$:



- The centrifugal force also admits a potential energy form in S_1 if $\boldsymbol{\omega}_{10}$ is constant:

$$V_1 = -\frac{1}{2} m^P (\omega_{10})^2 (r_{\perp})^2; \quad \mathbf{F}_C = -\nabla V_1 = m^P (\omega_{10})^2 \mathbf{r}_{\perp}$$

Inertia forces

- The Euler force $\mathbf{F}_E$ is always perpendicular to $\boldsymbol{\alpha}_{10}$ and $\mathbf{r}_1^P$
 - It does not admit a potential form ($\nabla \times \mathbf{F}_E \neq \mathbf{0}$)
- The Coriolis force $\mathbf{F}_{COR}$ is always perpendicular to $\boldsymbol{\omega}_{10}$ and $\mathbf{v}_1^P$. It is proportional to $\mathbf{v}_\perp$ (the component of $\mathbf{v}_1^P$ perpendicular to $\boldsymbol{\omega}_{10}$)

$\mathbf{F}_{COR}$ never
exerts work

Example: equatorial launch

- Launch toward the East

- Launch toward the West



Thank you!

